

MINGLANG YIN, Ph.D.

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AI · Predictive Modeling · Statistics · Multimodal · Uncertainty Quantification · High-Performance Computing

SUMMARY

ML/AI researcher, 10+ years modeling sparse, noisy, high-dimensional data. Depth in time-series forecasting, probabilistic modeling, survival analysis, and computational sciences. Hands on experience on HPC and distributed GPU training. Publications in Nature journals, AAAI, and top applied math & clinical journals. Seeking quantitative/AI research roles.

EDUCATION

Ph.D. (Biomedical Engineering) and M.Sc. (Fluid and Thermal Sciences) — Brown University 2016 – 2022

Division of Applied Mathematics · Advisor: Prof. George Em Karniadakis

B.S. (Aeronautical Engineering) — Northwestern Polytechnical University 2012 – 2016

RESEARCH EXPERIENCE

Senior Postdoctoral Researcher — Johns Hopkins University (JHU) Aug 2022 – Present

AI Lead @ ADVANCE, Advisors: Profs. Natalia Trayanova (BME/ADVANCE), Mauro Maggioni (Applied Math & Statistics)

Time-Series Foundation Model for Partial Differential Equations (PDEs) 2022 – present

- Proposed DIMON, a novel generic AI framework for predicting PDEs solution across a family of geometries. Achieved 10,000× speedup over numerical solvers, validated on 1000+ patients' hearts with over 98% accuracy.
- Developed a Time-Series Foundation Model and validated across 28+ interventions on 1000+ hearts with accuracy >95%.
- Reduced training time with distributed GPU training and Flash Attention; Improved generalizability with synthetic data.
- Published in *Nature Computational Science and conferences (AAAI 2024, SIAM CSE, ICME, and Heart Rhythm Scientific Sessions)*. Latest manuscript under review.

Multimodal AI for Classification and Survival Analysis with Calibration 2022 – present

- Developed a novel multimodal AI combining images with covariates for prognosis. Cross-validated on the internal/external data to test performance under distribution shift and led the deployment in external institutes.
- Proposed a new heteroscedastic non-proportional hazard model with multimodal AI for survival analysis. It outperformed established clinical scores, baseline Cox models, and unimodal AI.
- Designed and implemented the entire statistical analysis stack: hypothesis testing, feature selections, survival analysis, and model calibration. Improved explainability (SHAP & attn. rollout) and performance (loss for unbalanced data & ViT).
- Manuscript published in *Nature Cardiovascular Research* and *JACC: Clinical Electrophysiology*; accepted in *Journal of American Heart Association*; and presented in *Heart Rhythm Scientific Session*.

Representation Learning, Structural Feature Extraction, and Unsupervised Learning 2024 – present

- Developed a new dimension-reduction framework for analyzing high-dimensional geometric data. Outperformed baseline scores for risk forecast with selected shape features and yielded new quantitative clusters with novel scientific insights.
- Published in *Computers in Biology and Medicine*. Latest manuscript under review.

Ph.D. Researcher — Brown University, Division of Applied Mathematics 2018 – 2022

- **Bayesian inference & UQ**: Built a GAN-based generative model coupled with MCMC to infer parameters and quantify uncertainty from sparse, noisy data; benchmarked against physics-informed ML and Gaussian Process baselines.
- **High-Performance Computing**: implemented a custom particle solver for nonlinear mechanics within LAMMPS framework (not natively supported); implemented new boundary conditions in a parallel CFD solver; Co-designed a GPU–CPU parallel simulation pipeline; Built a molecular dynamics solver with CUDA.
- **Surrogate Modeling**: Developed physics-informed and data-driven surrogate models for nonlinear dynamical systems.
- **Publications**: *Journal of the Mechanics and Physics of Solids*, *Computer Methods in Applied Mechanics and Engineering*, *Journal of The Royal Society Interface*, *Computational Mechanics*, *AAAI*, etc.

SELECTED PUBLICATIONS

- **M. Yin et al.** Predicting the *When*: Multimodal AI for Time-to-Recurrence Analysis After Atrial Fibrillation Ablation, *Journal of American Heart Association*, 2026.
- C. Lai, **M. Yin et al.** *Multimodal AI to forecast arrhythmic death in hypertrophic cardiomyopathy*. *Nature Cardiovascular Research*, 2025. [[JHU Hub](#), [Johns Hopkins Medicine](#), [New York Magazine](#)]
- **M. Yin et al.** *A scalable framework for learning geometry-dependent solution operators of PDEs*. *Nature Computational Science*, 2024. [[99th percentile](#); [Communication of the ACM](#), [MIT Tech Review](#), [NSF News](#)]
- **M. Yin, et al.** *A generative modeling framework for inferring families of biomechanical constitutive laws in data-sparse regimes*. *Journal of the Mechanics and Physics of Solids*, 2023.

SELECTED GRANTS & AWARDS

- **AHA Rapid Impact Research Award** (Collaborator; based on my NIH K99 proposal) 2026
- **Best PhD Thesis Award** (Int. J. Numer. Methods Biomed. Eng.) 2024
- **Kenneth M. Rosen Fellowship** 2023
- **Chinese Government Award for Outstanding Students Abroad** Top 1/600 globally 2022

TECHNICAL SKILLS

Python, C++, CUDA, OpenMPI; PyTorch, TensorFlow; Distributed CPU/GPU computing; Numerical PDEs; Git, Linux; Time-series forecasting, generative models, Bayesian inference, MCMC, survival analysis, hypothesis testing, model calibration.